Task	HyenaDNA	Mamba-PS	Mamba-PH	xLSTM-PS	xLSTM-PH	*ours* GBST-BERT	Transformer++
Histone marks (MCC)							
H3	0.779 ± 0.037	0.799 ± 0.029	0.815 ± 0.048	0.796 ± 0.014	**0.824** ± 0.010	0.815 ± 0.031	0.801 ± 0.029
H3K14ac	0.612 ± 0.065	0.541 ± 0.212	0.631 ± 0.026	0.570 ± 0.008	0.598 ± 0.017	**0.673** ± 0.016	0.569 ± 0.062
H3K36me3	0.613 ± 0.041	0.609 ± 0.109	0.601 ± 0.129	0.588 ± 0.019	0.625 ± 0.010	**0.655** ± 0.035	0.591 ± 0.032
H3K4me1	0.512 ± 0.024	0.488 ± 0.102	0.523 ± 0.039	0.490 ± 0.012	0.526 ± 0.001	**0.565** ± 0.038	0.481 ± 0.037
H3K4me2	0.455 ± 0.095	0.388 ± 0.101	0.487 ± 0.170	0.489 ± 0.024	0.504 ± 0.012	**0.509** ± 0.212	0.367 ± 0.076
H3K4me3	0.549 ± 0.056	0.440 ± 0.202	0.544 ± 0.045	0.520 ± 0.019	0.537 ± 0.012	**0.553** ± 0.180	0.362 ± 0.219
H3K79me3	0.672 ± 0.048	0.676 ± 0.026	**0.697** ± 0.077	0.662 ± 0.011	**0.697** ± 0.007	**0.697** ± 0.132	0.676 ± 0.041
H3K9ac	0.581 ± 0.061	0.604 ± 0.048	0.622 ± 0.030	0.622 ± 0.013	0.627 ± 0.008	**0.648** ± 0.037	0.599 ± 0.046
H4	0.763 ± 0.044	0.789 ± 0.020	0.811 ± 0.022	0.793 ± 0.011	**0.813** ± 0.008	0.804 ± 0.074	0.803 ± 0.022
H4ac	0.564 ± 0.038	0.525 ± 0.240	0.621 ± 0.054	0.558 ± 0.018	0.583 ± 0.014	**0.642** ± 0.037	0.528 ± 0.036
Enhancers (MCC)							
enhancers	0.517 ± 0.117	0.491 ± 0.066	**0.546** ± 0.073	0.375 ± 0.030	0.545 ± 0.024	0.336 ± 0.085	0.368 ± 0.074
enhancers_types	0.386 ± 0.185	0.416 ± 0.095	0.439 ± 0.054	0.444 ± 0.046	**0.466** ± 0.011	0.430 ± 0.167	0.384 ± 0.053
Promoters (F1)							
promoter_all	0.960 ± 0.005	0.967 ± 0.004	0.970 ± 0.004	0.962 ± 0.002	0.967 ± 0.001	**0.971** ± 0.003	0.968 ± 0.006
promoter_no_tata	0.959 ± 0.011	0.968 ± 0.006	0.968 ± 0.010	0.963 ± 0.002	0.970 ± 0.001	**0.972** ± 0.006	0.969 ± 0.004
promoter_tata	0.944 ± 0.040	**0.957** ± 0.015	0.953 ± 0.016	0.948 ± 0.006	0.952 ± 0.005	0.948 ± 0.021	0.953 ± 0.013
Splice sites (accuracy / F1)							
splice_sites_all	0.956 ± 0.011	0.927 ± 0.021	0.940 ± 0.027	0.965 ± 0.006	0.974 ± 0.004	**0.980** ± 0.004	0.975 ± 0.006
splice_sites_acceptors	0.958 ± 0.010	0.936 ± 0.077	0.937 ± 0.033	0.970 ± 0.005	0.953 ± 0.008	**0.977** ± 0.009	0.955 ± 0.049
splice_sites_donors	0.949 ± 0.024	0.874 ± 0.289	0.948 ± 0.025	0.962 ± 0.004	0.951 ± 0.005	**0.980** ± 0.012	0.909 ± 0.117
Tasks with the best mean	**0**	**1**	**2**	**0**	**4**	**13**	**0**

Mean test score over 10 repeats (higher is better); small number = range (best minus worst repeat). Highlighted = best mean for that task (ties counted for each model). First five columns: Bio-xLSTM paper, Table 1.